

orientation, acceleration, pressure, temperature, and servo positions. The same telemetry data is simultaneously transmitted through the UART interface (TX: GPIO 43 and RX: GPIO 44), allowing an external radio telemetry module to be connected for long-range communication with a ground station during flight.

In addition, the firmware logs the flight parameters throughout the flight. After recovery using a parachute, the recorded data can be analysed to evaluate the rocket's performance and further optimise the design, PID parameters, and flight characteristics for subsequent launches.

EFY note. Proper PID tuning ensures stable flight by

compensating for wind, turbulence, and other disturbances.

Software/Coding

The Arduino IDE is used to develop, compile, and upload the flight controller firmware. Before compiling the code, install the required BME280 and MPU6050 libraries in the Arduino IDE. After completing the firmware, select the correct board and COM port in the Arduino IDE, then compile and upload the firmware to the IndusBoard Coin. Ensure that all sensors and servo motors are functioning correctly before proceeding with flight testing.

Assembly

Begin by assembling the rocket body, including the body tube, nose cone, fins, launch lug, and other mechanical components.

Next, prepare the rocket motor and igniter. Carefully install the motor mount at the bottom of the rocket body tube and ensure that it is firmly secured. Insert the Estes C6-5 model rocket engine, or the engine selected for the design, into the motor mount. Then insert the igniter into the nozzle until its tip makes contact with the propellant inside the motor. Secure the igniter in place using the supplied plastic igniter plug.

Once the motor and igniter have been prepared, slide the motor fully into the motor mount and lock it securely. Route the igniter leads so that they remain easily accessible for connection to the launch controller or the onboard ignition circuit.

See Fig. 4 and Fig. 5 for the assembly procedure. Fig. 4 shows the model rocket motor and igniter, while Fig. 5 shows the rocket motor installed in the motor mount together with the igniter and the electronics wiring.

Preparing the servo-controlled fins

Next, prepare the servo-controlled fins for the stabilisation system. Each fin is attached to a micro servo motor, which is mounted in the lower

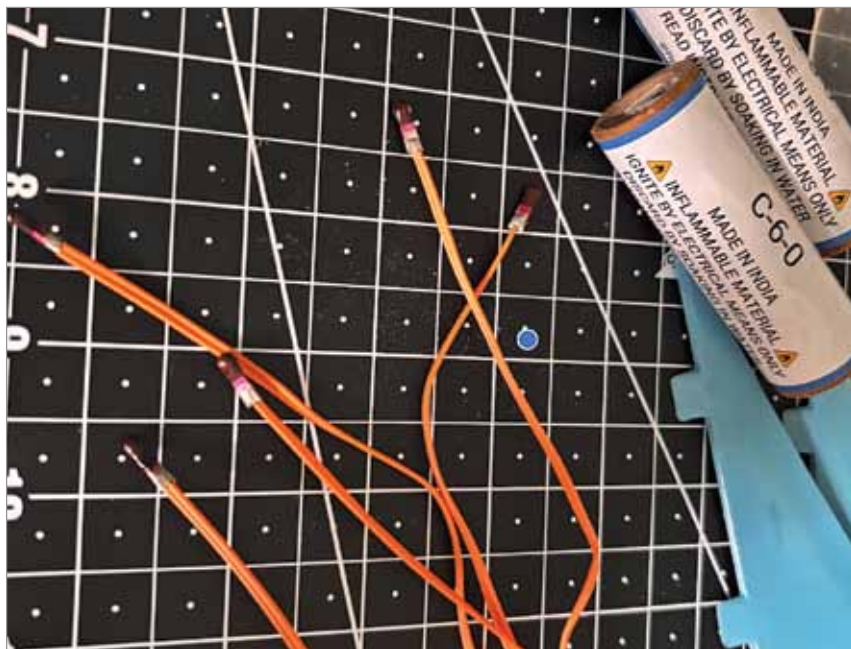


Fig. 4: Model rocket motor and igniters



Fig. 5: Rocket motor installed with the igniter and electronics wiring